

Erdős Problem #627

EP-627

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $\omega(G)$ denote the clique number of G and $\chi(G)$ the chromatic number. If $f(n)$ is the maximum value of $\chi(G)/\omega(G)$, as G ranges over all graphs on n vertices, then does $\lim_{n \rightarrow \infty} \frac{f(n)}{n/(\log n)^2}$ exist?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>